

Ashish Kumar

(Publishing as Ashish S. Kumar)

Ph.D. Scholar (Thesis Submitted)

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Google Scholar

Gandhinagar, India

Nationality: Indian

Research Interests

Urban and riverine flood risk modelling; flood adaptation and risk redistribution; equitable urban resilience; hydrologic and hydrodynamic modelling; flood–structure interactions; nature-based and hybrid adaptation; climate-risk assessment; physics-guided and data-driven flood modelling; heat risk; Earth observation and remote sensing for environmental analysis; geospatial decision support; translating flood and climate-risk research into actionable insights for adaptation planning.

Education

Ph.D. in Civil Engineering

July 2021–Present

Indian Institute of Technology Gandhinagar, India

CPI: 9.50/10

Thesis Title: *Moving Flood Adaptation beyond Isolated Protection towards Connected and Equitable Risk Management across Spatial Scales.*

Supervisor: Dr. Udit Bhatia.

Thesis submitted: 14 August 2026; defence expected: October 2026.

B.Tech. in Civil Engineering

2016–2020

National Institute of Technology Jamshedpur, India

CGPA: 7.97/10.

Publications

Journal Publications

- **Kumar, A. S.**, Majumder, R., Kapadia, V. P., and Bhatia, U. (2025). Partial flood defenses shift risks and amplify inequality in a core–periphery city. *Nature Cities*, 2, 835–846. <https://doi.org/10.1038/s44284-025-00299-7>.
- **Kumar, A. S.**, Pandey, A. K., Mohapatra, P. K., and Bhatia, U. (2025). Three-dimensional numerical study of flood–building interactions to assess the efficiency of flood adaptation strategies. *Journal of Hydraulic Engineering*, 151(5), 04025028. <https://doi.org/10.1061/JHEND8.HYENG-14299>.
- Borah, A., Datta, A., **Kumar, A. S.**, Dave, R., and Bhatia, U. (2026). Dense canopies reverse the cooling effect of urban greening in humid cities. *Nature Communications*, 17, 5997. <https://doi.org/10.1038/s41467-026-72636-w>.
- Poonia, H., Upadhyay, D., Nikumbh, A. C., **Kumar, A. S.**, Murtugudde, R., and Bhatia, U. Internal climate variability reshapes global extreme-rainfall synchronization networks. *npj Climate and Atmospheric Science*. Under review.

Patent Applications

- **Kumar, A.** and Bhatia, U. (2025). Protection-Induced Time-Shift Method for Quantifying Temporal Redistribution of Urban Flood Risk. Indian Patent Application No. **202521103502**, filed on 27 October 2025.
- **Kumar, A.** and Bhatia, U. (2026). Dual Barcode Method for Quantifying Spatial Redistribution of Dry Land under Flood Protection. Indian Patent Application No. **202621008727**, filed on 28 January 2026.

Conference Presentations

- **Kumar, A.**, Majumder, R., Kapadia, V. P., and Bhatia, U. (2026). Risk redistribution and inequality under partial flood protection in a core–periphery city. *EGU General Assembly*, EGU26-18151. <https://doi.org/10.5194/egusphere-egu26-18151>. **Poster presentation.**
- **Kumar, A.** and Bhatia, U. (2025). Flood risk redistribution due to gaps and constraints in adaptation strategies. *EGU General Assembly*, EGU25-15089. <https://doi.org/10.5194/egusphere-egu25-15089>. **Oral presentation.**
- **Kumar, A.**, Datta, A., and Bhatia, U. (2025). How Creek Widening and Wetland Buffers Flip Urban

Flood Hazards to Protection. *AGU Fall Meeting 2025*, New Orleans, LA, NH43D-048. <https://ui.adsabs.harvard.edu/abs/2025AGUFMNH43D.048K>. **Poster presentation.**

- **Kumar, A.** and Bhatia, U. (2024). Integrating shared risk into the economic assessment of flood adaptation. *AGU Annual Meeting*. <https://doi.org/10.22541/essoar.174426927.77603029/v1>. **Oral presentation.**
- **Kumar, A.** and Bhatia, U. (2024). Evaluating flood adaptation effectiveness through economic analysis: A case study of Surat, India. *EGU General Assembly*, EGU24-7335. <https://doi.org/10.5194/egusphere-egu24-7335>. **Poster presentation.**
- Datta, A., **Kumar, A.**, Dubey, S., and Bhatia, U. (2025). Dynamic Management Strategies for Plant Pollinator Networks under Anthropogenic Warming Scenarios. *EGU General Assembly*, EGU25-779. <https://doi.org/10.5194/egusphere-egu25-779>.
- Sutrave, M., **Kumar, A.**, Dave, R., and Bhatia, U. (2025). Quantifying Subsurface Contributions to Compound Flooding in Coastal Urban Areas for Enhanced Resilience. *EGU General Assembly*, EGU25-867. <https://doi.org/10.5194/egusphere-egu25-867>.

Field Experience

- **Chaliyar and Kallayi River Bathymetry Survey, Kozhikode, Kerala:** Conducted ADCP- and RTK-based bathymetric surveys, generated river cross-sections, collected information on the depth and extent of the 2018 flood, and developed a preliminary riverine flood model using the surveyed cross-sections.
- **Railway Underpass Water-Seepage Assessment, Gondal, Gujarat:** Investigated chronic water seepage and underpass flooding associated with adjacent reservoirs and surface runoff, delineated the interacting groundwater-surface-water system, and developed a modelling strategy to support mitigation planning.
- **Flood Data Collection and Site Reconnaissance, Surat, Gujarat:** Collected hydrological, hydraulic, reservoir-operation, and historical flood data and conducted city-wide reconnaissance to understand flood pathways and reconstruct the behaviour of the 2006 flood event.
- **Narmada River Intake-Well Survey, Jhagadia, Gujarat:** Conducted an ADCP-based bathymetric survey and mapped riverbed topography near a river intake structure to support water-resource assessment.

Teaching Experience

- **Graduate Teaching Fellow, Hydrology and Hydraulics, IIT Gandhinagar** Jul–Nov 2025
Designed and taught laboratory modules in hydrology, hydraulic measurements, hydrological and hydrodynamic modelling, and experimental flume analysis.
- **Teaching Assistant, Networks and Complex Systems, IIT Gandhinagar** 2022–2024
Guided interdisciplinary course projects and supported student evaluation.
- **Teaching Assistant, Infrastructure Systems, Planning and Management, IIT Gandhinagar** 2023
Guided projects on infrastructure planning, flood modelling, risk assessment, and resilience.
- **PMRF Teaching Fellow, Remote Sensing using Google Earth Engine, L.D. College of Engineering** Mar–May 2024
Delivered lectures and hands-on laboratory sessions in remote sensing and Google Earth Engine.

Mentoring Experience

- **Master's Thesis Mentoring:** Mentored three Master's thesis projects on subsurface contributions to compound urban flooding, flood fragility of low-rise buildings, and groundwater-flood mitigation using pumping and underground drainage wells.
- **Undergraduate Project Mentoring:** Mentored projects on AI/ML-assisted flood prediction using physics-based modelling and sensor-based real-time urban flood-depth monitoring.

Technical and Communication Skills

- **GIS, Earth Observation, and Remote Sensing:** ArcGIS, QGIS, and Google Earth Engine; DEM processing, raster/vector analysis, flood-hazard and exposure mapping, and spatial-data analysis.
- **Hydrologic and Hydraulic Modelling:** MIKE+ 1D–2D, HEC-RAS, ANUGA, HEC-HMS, SWMM, and SWAT; experience in riverine, pluvial, urban, and dam-break flood modelling.
- **Computational and Analytical Skills:** ANSYS Fluent, Python, MATLAB, C++, and STATA; 3-D CFD, flood-structure interaction, large model-output processing, time-series analysis, uncertainty analysis, economic-impact assessment, and machine learning.
- **Scientific Communication:** Scientific writing and presentation through peer-reviewed publications, tech-

nical reports, and oral and poster presentations at international conferences including EGU and AGU.

- **Field Skills:** ADCP- and RTK-based bathymetric surveys, river cross-section development, flood-depth and inundation surveys, field reconnaissance, and data collection.
- **Teamwork and Collaboration:** Interdisciplinary research involving hydrology, hydraulics, urban climate, remote sensing, and engineering, resulting in collaborative publications including *Nature Communications*.
- **Engineering Software:** AutoCAD and STAAD.Pro.

Honours and Awards

- **Prime Minister's Research Fellowship (PMRF)**, Government of India 1 Aug 2023–16 Jul 2026
- **International Travel Grant**, Anusandhan National Research Foundation (ANRF), Government of India 2025

Academic Service and Extracurricular Activities

- **Session Co-Convener**, EGU General Assembly 2026, Session BG10.12, “From Urban Heat to Flood Risk: Integrating Geospatial Data, Models, and Observations for Green–Blue Adaptation Strategies.”
- **Peer Reviewer** for international journals, including *Journal of Hydrology* and *Nature Water*.
- **Extracurricular Interests:** Playing badminton, advancing my swimming skills, and exploring different Indian regional cuisines through cooking.